REVIEW ARTICLE

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Intraperitoneal local anaesthetic in abdominal surgery – a systematic review

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intraperitoneal, local anaesthetic, pain, surgery, viscera.

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Abstract

Background: The use of intraperitoneal local anaesthetic (IPLA) can be used to modulate visceral nociception after abdominal surgery; however, this technique is not routinely used in open abdominal surgery. The aim of this systematic review was to appraise the clinical effects of IPLA in open abdominal surgery for metachronous outcomes including pain, metabolic response to surgery and gastrointestinal function.
Methods: A comprehensive search was conducted independently without language restriction. Relevant meeting abstracts and reference lists were manually searched. Data analysis was performed using Review Manager Version 5.0 software. Post-operative clinical and metabolic outcomes of randomized controlled trials comparing IPLA versus no IPLA or placebo solution were used for meta-analysis.
Results: Twelve trials were identified including eight randomized trials in gastrointestinal and gynaecological surgery. Post-operative pain was reduced but not opioid use. There was blunting of postoperative hyperglycaemia. There was no difference in post-operative cortisol response. Return of bowel function appeared to be quickened, although meta-analysis was not possible.
Conclusion: The use of IPLA is safe and appears to have clinical benefits. However this technique has not been studied in optimized perioperative settings. Trials are needed to evaluate this method of visceral blockade further after major abdominal surgery.

Introduction

Compared with somatic pain, visceral nervous activity is a distinctly separate form of nociception.[1] In addition so called 'silent nociceptors' are activated by intraperitoneal (IP) inflammation and injury as seen after surgery.[1] The vagal afferent responses gives rise to both painful and non-painful sensations and influenced feeding and illness behaviours.[2,3] With large peritoneal injury or inflammation, as seen after abdominal surgery, these effects are thought to contribute to organ dysfunction and morbidity.[4–6] This peritoneal activation maybe central to understanding the prolonged post-operative course that patients commonly suffer after open abdominal surgery that is not seen limb surgery of the same magnitude.[7]

Somatic pain arising from the surgical injury of the abdominal wall can be controlled by epidural blockade.[8] It is also possible to instil local anaesthetic (LA) solutions into the peritoneal cavity thereby blocking visceral afferent signalling, and further modifying pain and illness responses. Intraperitoneal local anaesthetic (IPLA) have been used as 'visceral blocks' since as early as 1950[9,10] and IPLA has been successfully used in laparoscopic surgery, to reduce shoulder tip pain, overall pain, nausea and vomiting and hospital stay.[11–16] This model of complete 'noci- association' of the surgical stress forms a basis of Enhanced Recovery After Surgery or ‘fast-track’, advances in perioperative care.[17]

IP administration of LA is not routine in modern day open abdominal surgery. This is perhaps due to the lack of attention paid to the concept of ‘visceral nociception’ as a significant source of post-operative morbidity and impressive advances in somatic analgesia techniques. The aim of this review was to systematically evaluate the available information on the use of IPLA agents in open abdominal surgery, focusing on clinical outcomes as indicators of the effectiveness of the reduction of visceral nociception by this modality.

Methods

Search strategy

A comprehensive database search was carried out independently by the first two authors (AK, TS). The following databases were searched from inception to August 2009: Medline, CINHAL, Pubmed, Cochrane Controlled Trials Register and EMBASE. The search terms used were: ‘local anaesth$’, ‘local anesth$’, ‘nerve conduct$’, ‘Ropivicaine’, ‘Bupivacaine’, ‘Lidocaine’, ‘Lignocaine’, ‘Procaine’, ‘Intra-abdominal$’, ‘intraperitoneal’, ‘visceral’, ‘laparosc$’, ‘peritoneum’, ‘coeliosco$’, ‘surgery’. Language limitation was not used. Additional papers or abstracts were retrieved by hyperlinks and by manually scrutinizing the reference lists of relevant publications. Animal trials were excluded. Academic meeting proceedings were also searched by the second author. Publications in languages other than English were translated into English with the help of academics at the Faculty of Medical and Health Sciences at the University of Auckland. Authors were contacted for additional data when required.

Study selection

Papers were selected for the review if they reported intra-abdominal LA use for any indication pre, intra or post-operatively. Exclusion criteria included trials using laparoscopy, trials evaluating the use of pre-peritoneal or abdominal wall (incisional) LAs (unless IPLA was in the comparison). All randomized controlled trials (RCTs) and nonrandomized controlled comparison trials (CCTs) were included for review. However, only double-blinded RCTs were used for meta-analysis. AK (blinded) and TS independently examined all retrieved papers and any disagreement over inclusion or exclusion was discussed with the senior author (AGH) and a consensus reached. Methodological quality of RCTs was assessed using the previously published Jadad criteria.[18]

Data extraction

The primary outcomes used for quantitative analysis were pain scores, opioid analgesia requirement and measures of neuroendocrine response and adverse events. Any other outcomes relating to recovery such as gastrointestinal function, respiratory outcomes and time to discharge were also considered for quantitative analysis if measured uniformly. Data extraction was performed and entered into tables based on the following post-operative time points: 1, 2, 4, 6, 8 and 24 h. The median score was used as an estimate of mean where the latter was not reported. Standard deviation measures were obtained from corresponding author if not published, or calculated based on the methods described in the Cochrane Handbook of Systematic Reviews of Interventions.[19] Only pain scores reported using the visual analogue scale (VAS, 0–100 mm or 0–10 cm) were used for analysis. The pain score that best represented abdominal pain at rest was used. In addition, pain outcomes were only analysed by meta-analysis if an IPLA arm was compared with an IP placebo solution or a ‘no treatment’ arm. If additional modalities were tested as treatment, then these groups were excluded from meta-analysis. Total opioid analgesia requirement within 24 h was extracted from the reports where possible and converted into morphine equivalence.[20] Non-steroidal analgesia was not converted into morphine equivalence and not included in meta-analysis. Neuroendocrine measures were similarly obtained and categorized based on time interval post-operatively and converted into International System of Units (S.I.) units before being used for meta-analysis.

Statistical analysis

Publication bias was tested using the funnel plot inspection method. Meta-analysis was performed using Review Manager Version 5.0 (Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2008). Results of the meta-analysis were assessed by graphical presentations of mean or standardized mean difference according to the outcome and 95% confidence intervals represented on forest plots with $P < 0.050$ considered statistically significant. A chi-squared test for heterogeneity was performed and $P < 0.100$ was considered statistically significant for this. If significant heterogeneity existed, a sensitivity analysis to detect any small study effect was then carried out by comparison of the fixed and random effects, as detailed in the Cochrane Handbook of Systematic Reviews of Interventions.[19] Study weight was by sample size.

Results

The Quality of Reporting of Meta-analyses[21] (QUOROM) diagram is presented (Fig. 1). Twelve trials were identified in this review (Table 1). There were four CCTs and eight RCTs. Quality of RCTs varied from 2–5 based on the Jadad criteria. There were no adverse events or intra-abdominal infections reported in any of these studies. The trials reported various procedures including gastrectomy, cholecystectomy, acute laparotomy, appendicectomy and abdominal hysterectomy. Various methods of administration such as IP infusion catheters, splanchnic plexus injection and free instillation were reported.

Safety

There were no reports of adverse events after IPLA use including systemic toxicity or infective complications.

Pain by VAS

Pain was assessed in seven RCTs.[34–40] However, only six trials compared IPLA with placebo or no IPLA as one trial was a dose finding study.[40] There was a significant reduction of pain as measured on VAS at 1, 8 and 24 h in favour of IPLA in 595 patients (overall standardized mean difference (SMD) −0.38 (−0.54, −0.21); P <

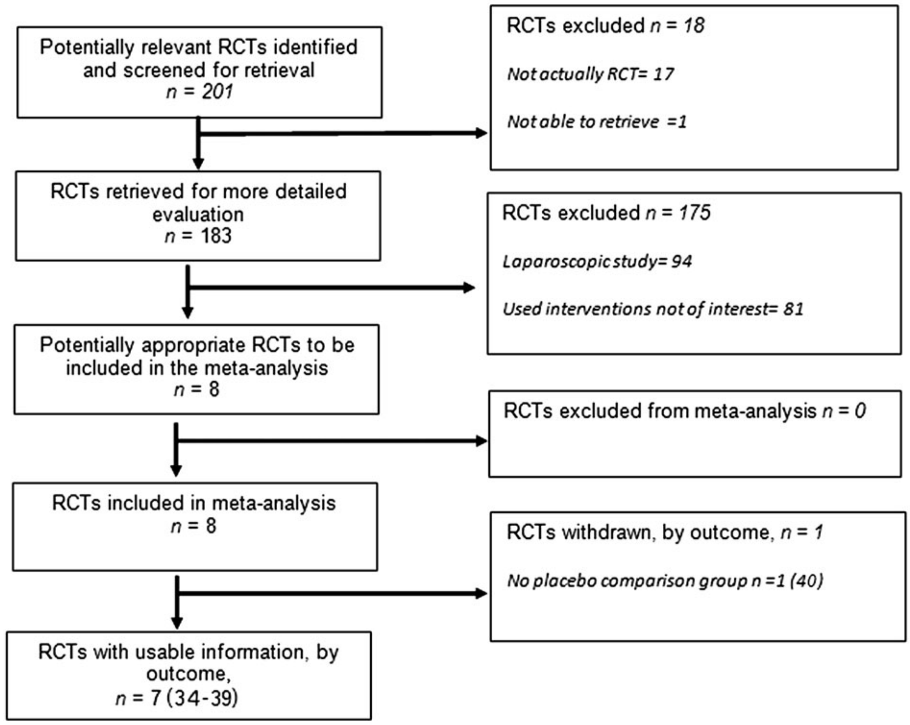

Fig. 1. Quality of Reporting of Meta-analyses[21] (QUOROM) diagram.

0.00001) (Fig. 2). The funnel plot was symmetrical indicating no reporting bias and overall heterogeneity was not present.

Morphine usage

Opioid analgesia use within the first 24 h after surgery was converted and equated using morphine equivalence conversion.[20] Two hundred eighteen patients were studied in six RCTs in a random effects model.[34–39] This revealed that there was no significant difference (overall SMD –0.13 (–0.68, 0.42); $P = 0.64$) (Fig. 3); however, there was significant heterogeneity, likely due to the different perioperative protocols and agents used in these RCTs.

Gastrointestinal function

Three CCTs and one RCT reported a measure of recovery of gastrointestinal function in 320 patients.[27–29,41] These measures included using electroenterogram readings[27–29] and radio-opaque markers.[41] Three trials reported quicker return of bowel motility and one trial reported a significant reduction for the need of re-operations for adhesiolysis during follow-up. Meta-analysis could not be performed due to various modes of measuring bowel motility as well as the lack of RCTs in this area.

Metabolic outcomes

Four RCTs in 209 patients reported measures of post-operative metabolic measures at various time points.[30,34,35,41] Using serum glucose as a marker, all RCTs reported some attenuation of post-operative hyperglycaemia and meta-analysis of RCTs confirmed this finding at 1 and 4 h but not at 8 h post-operatively (overall SMD –0.87 (–1.44, –0.30); $P = 0.003$) (Fig. 4). The funnel plot showed no asymmetry and study sizes were similar; however, there was significant heterogeneity. Therefore, fixed and random-models were compared and the result of the meta-analysis did not alter (stayed in favour of IPLA).

Three trials measured serum cortisol post-operatively in 143 patients.[30,34,35] Meta-analysis revealed that there was no significant reduction (overall SMD –0.00 (–0.63, 0.62); $P = 0.99$). In fact, serum cortisol was significantly higher in the IPLA groups at 4 h (Fig. 5). The funnel plot was symmetrical. However, heterogeneity was present although the result of meta-analysis did not change after sensitivity analysis.

Discussion

This is the first meta-analysis investigating IPLA in open abdominal procedures. In this systemic review we identified 12 trials including eight RCTs for gastrointestinal and gynecological indications. The methods of administration, types of LA, duration of administration and dosages were varied between studies. There were no reports of adverse effects in any patients, including IP sepsis from infusion catheters or systemic toxicity from LAs. Meta-analysis of pain outcomes revealed a reduction of pain but not opioid use with IPLA. Meta-analysis detected a reduction in the post-operative serum glucose response with IPLA but not with cortisol. Several trials demonstrated quicker return of bowel function after open surgery as measured by radio-opaque markers and electroenterograms, but unfortunately, meta-analysis was not possible due to the differences in measurements used.[28,29,41]

One paper identified in this review was a dose finding study.[40] The authors showed that after hysterectomy, 7.5, 12.5 and 17.5 mg/h of levobupivacaine (at 10 mL/h) all resulted in similar analgesia. Based on these data, it is logical that the lowest dose infusion of 7.5 mg/h

Table 1 Trials in open general surgical procedures

Reference, first author, year,	Trial method, procedure, N (intervention/control groups)	Timing of LA use: Pre or post with respect to dissection, method of delivery	Agent used, comparison group	Pain outcome, analgesia use	Gastrointestinal motility outcomes	Main findings in intervention group	Jadad score, used for meta-analysis?
[29] Zhitunik, 1969	CCT, open gastric resection, 30/20	Post, infusion catheter placed over small bowel	500–600 mL 0.25% procaine bolus then 200–300 mL 0.25% infusion system over next 3–4 days, controls	NA	Electro-gastrograph measures of bowel peristalsis.	Reduced post op hypokinesia	Excluded as non-randomized CCT
[30] Tsuji,1983	CCT, gastrectomy 12/14/14	Post, splanchnic plexus infiltration in addition to epidural	45 mL 0.5% bupivacaine injected directly into splanchnic plexus in addition to epidural, GA alone, epidural alone	NA	NA	Reduced endocrine response including cortisol, glucose and abolished free fatty acid response in splanchnic blockade group. No side effects.	Excluded as non-randomized CCT
[27] Tishinskaia, 1984	CCT, laparotomy for adhesive bowel obstruction 51/61	Post, instillation then post op intermittent infusion system via IP catheter	0.25% procaine solution bolus at end of op, then q8 h of 4–5 mL/kg 0.25% procaine for 3–5 days post op, controls	NA	Reduced number of re-operations in 1–13 years of follow-up	Intra-abdominal infusion of procaine improves long term outcome with reduced recurrence after adhesion surgery	Excluded as non-randomized CCT
[28] Sandler, 1984	CCT, laparotomy for acute general procedures 32/32/32	Post, via infusion catheter placed at the root of small intestine mesentry	150–200 mL 0.25% procaine three times daily for first post op day, stimulation therapy, drug induced sympathetic block	NA	Quicker return of peristalsis, passage of flatus, stool and greater frequency and amplitude on the electroenterogram	IPLA speeds recovery of intestine function	Excluded as non-randomized CCT
[22] Rimback, 1986	RCT, open cholecystectomy or gastric procedures 11/11	Pre op intraabdo instillation via right subcostal incision	2 mg/kg IP bupivacaine in 300 mL NS, NS	NA	Shortened colonic inhibition as measured using swallowed markers and serial x-rays	Safe, reduced hyperglycaemic response to surgery.	4, Used for glucose meta-analysis
[35] Wallin, 1988	RCT, open cholecystectomy 9/10	Pre op intraperitoneal via right subcostal incision.	2 mg/kg IP bupivacaine in 300 mL NS v NS	No significant difference in pain or pethidine use	NA	Decreased hyperglycaemic response, no change in pain or serum cortisol levels.	2, Used for pain, pethidine, glucose, cortisol meta-analysis.
[34] Scott, 1988	RCT, open upper and lower GI procedures (not specified) 10/10	Pre-dissection instillation, post op infusion via catheter under right crus of diaphragm or aortic bifurcation	1 mg/kg bupivacaine in 300 mL NS bolus, pre-dissection, post-op infusion of 20 mL/hr 0.25% bupivicaine for 8 h, NS	No significant difference in pain or morphine use	NA	No difference in post operative glucose, cortisol or respiratory outcomes.	3, Used for pain, morphine, glucose and cortisol.
[36] Williamson 1997	RCT, total abdominal hysterectomy	Post, instilled into pelvic cavity after hemostasis.	50 mL 0.4% lignocaine solution with 1:500 000 adrenaline, NS.	Reduced visual analogue pain at rest, no difference in opioid consumption	No difference in PONV	Safe, modest reduction in pain.	3, Used for pain and morphine use
[37] Ali 1998	RCT, total abdominal hysterectomy	Post, instilled into pelvic cavity	50 mL 0.8% lignocaine with 1:500 000 adrenaline, 50 mL 0.2% bupivacaine with 1:500 000 adrenaline, NS	No difference on pain or morphine use	NA	No appreciable gain after abdominal hysterectomy	4, used for pain and morphine use
[38] Colbert, 1998	RCT, open appendicectomy 32/28	Post, direct peritoneal infiltration with bleb formation	1.5 mg/kg bupiv IP injection and SC, SC alone	Reduced pain, longer time to analgesia, reduced diclofenac and meperidine use	NA	Safe, improved pain, merits widespread use. Measures serum levels.	3, Used for pain and meperidine use
[39] Gupta 2004	RCT, open abdominal hysterectomy	Post, infused via multihole catheter placed supravaginally and inserted percutaneously	0.25% levobupicaine infused via catheter for 24 h, NS	Reduced early pain and 24 h opioid use	No difference in PONV or in any other recovery parameters	Reduced opioid use over first 24 h (infusion period).	4, used for pain and opioid use
[40] Perniola 2009	RCT, open abdominal hysterectomy	Post, infused via multihole catheter placed supravaginally	7.5 mg/h, 12.5 mg/h or 17.5 mg/h of levobupivacaine infused over 48 h.	No difference in pain or analgesia use	No difference in PON, no difference in respiratory and recovery parameters.	No dose dependant effect was found with lower dose infusion being just as effective as higher dose.	5, excluded as there was no comparison group with placebo or no intervention

CCT, comparative clinical trial; RCT, randomized clinical trial; IP, intraperitoneal; NS, normal saline; FFA, free fatty acids; PONV, post operative nausea and vomiting; LA, local anaesthetic; NA, not applicable; GI, gastrointestinal.

	Local anesthetic			Control				Std. Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Fixed, 95% CI
3.1.1 Pain 1hr								
Colbert 1998	3.2	1.1	32	5.4	2.1	28	8.6%	-1.32 [-1.88, -0.76]
Gupta 2004	3.4	2.9	20	5.4	2.8	20	6.7%	-0.69 [-1.33, -0.05]
Scott 1988	5.7	1	10	5.1	0.8	10	3.3%	0.63 [-0.27, 1.54]
Wallin 1988	4.5	2.1	9	3	3.2	10	3.2%	0.52 [-0.40, 1.44]
Subtotal (95% CI)			**71**			**68**	**21.8%**	**-0.56 [-0.91, -0.20]**
Heterogeneity: Chi² = 19.22, df = 3 (P = 0.0002); I² = 84%								
Test for overall effect: Z = 3.08 (P = 0.002)								
3.1.2 Pain 4hr								
Gupta 2004	1.5	1.7	20	1.6	1.6	20	7.1%	-0.06 [-0.68, 0.56]
Scott 1988	4.8	1.1	10	5	1	10	3.5%	-0.18 [-1.06, 0.70]
Wallin 1988	3.3	3.2	9	4.2	2	10	3.3%	-0.33 [-1.23, 0.58]
Subtotal (95% CI)			**39**			**40**	**13.9%**	**-0.15 [-0.60, 0.29]**
Heterogeneity: Chi² = 0.23, df = 2 (P = 0.89); I² = 0%								
Test for overall effect: Z = 0.68 (P = 0.50)								
3.1.3 Pain 8hr								
Ali 1998	2.6	1.8	20	3.6	2.4	20	6.9%	-0.46 [-1.09, 0.17]
Colbert 1998	3.6	2.3	32	4.3	2.1	28	10.5%	-0.31 [-0.82, 0.20]
Gupta 2004	2.5	1.8	20	2.7	2.3	20	7.1%	-0.09 [-0.72, 0.53]
Scott 1988	3	1	10	4.7	1	10	2.5%	-1.63 [-2.67, -0.59]
Wallin 1988	1.8	2.9	9	2.9	3.4	10	3.3%	-0.33 [-1.24, 0.58]
Subtotal (95% CI)			**91**			**88**	**30.3%**	**-0.41 [-0.71, -0.11]**
Heterogeneity: Chi² = 6.44, df = 4 (P = 0.17); I² = 38%								
Test for overall effect: Z = 2.66 (P = 0.008)								
3.1.4 Pain 24hr								
Ali 1998	1.9	1.5	20	2.5	2.1	20	7.0%	-0.32 [-0.95, 0.30]
Colbert 1998	2.3	1.7	32	3.4	1.6	28	10.0%	-0.66 [-1.18, -0.13]
Gupta 2004	2.8	2.2	20	2.5	2.2	20	7.1%	0.13 [-0.49, 0.75]
Wallin 1988	2.2	5.4	9	1	1.6	10	3.3%	0.30 [-0.61, 1.20]
Williamson 1997	2.6	1.8	20	3.9	2.1	19	6.5%	-0.65 [-1.30, -0.01]
Subtotal (95% CI)			**101**			**97**	**34.0%**	**-0.33 [-0.61, -0.05]**
Heterogeneity: Chi² = 6.43, df = 4 (P = 0.17); I² = 38%								
Test for overall effect: Z = 2.28 (P = 0.02)								
Total (95% CI)			**302**			**293**	**100.0%**	**-0.38 [-0.54, -0.21]**
Heterogeneity: Chi² = 34.44, df = 16 (P = 0.005); I² = 54%								
Test for overall effect: Z = 4.48 (P < 0.00001)								
Test for subgroup differences: Chi² = 2.11, df = 3 (P = 0.55), I² = 0%								

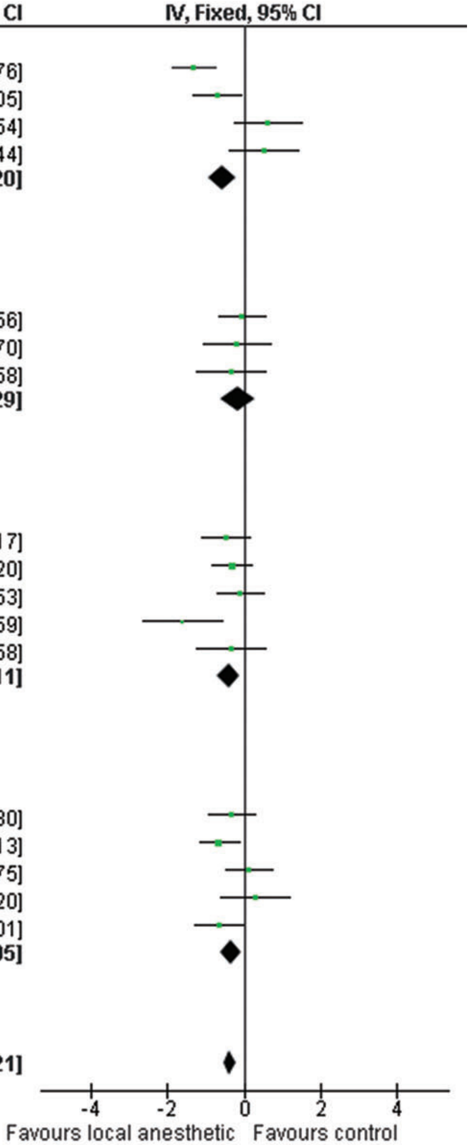

Fig. 2. Meta-analysis of the effect of intraperitoneal local anaesthetic on post-operative pain in all open general surgical procedures.

	Local anesthetic			Control				Std. Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Ali 1998	52.5	23.7	20	54.6	26.4	20	17.9%	-0.08 [-0.70, 0.54]
Colbert 1998	6.8	2.5	32	10	3.8	28	18.9%	-1.00 [-1.54, -0.46]
Gupta 2004	30.5	15.4	20	44.3	21.6	20	17.6%	-0.72 [-1.36, -0.08]
Scott 1988	16.4	7.6	10	10.4	4.4	10	13.9%	0.93 [-0.01, 1.86]
Wallin 1988	22.8	14	9	15	13.8	10	14.0%	0.54 [-0.38, 1.46]
Williamson 1997	59.5	19.2	20	59.8	15.7	19	17.8%	-0.02 [-0.64, 0.61]
Total (95% CI)			**111**			**107**	**100.0%**	**-0.13 [-0.68, 0.42]**
Heterogeneity: Tau² = 0.34; Chi² = 19.07, df = 5 (P = 0.002); I² = 74%								
Test for overall effect: Z = 0.46 (P = 0.64)								

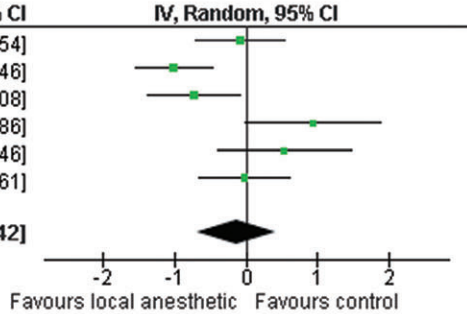

Fig. 3. Meta-analysis of the effect of intraperitoneal local anaesthetic on post-operative opioid use in all open general surgical procedures.

Study or Subgroup	Local anesthetic Mean	SD	Total	Control Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.6.1 Glucose 1hr post-op								
Rimback 1986	4.5	0.8	11	6	0.6	11	9.4%	-2.04 [-3.11, -0.97]
Scott 1988	8	0.5	10	8.2	0.4	10	10.5%	-0.42 [-1.31, 0.47]
Tsuji 1983	5.8	1.4	12	6.9	0.7	14	10.8%	-0.99 [-1.81, -0.16]
Wallin 1988	5.5	1.2	9	6.4	0.6	10	10.0%	-0.92 [-1.88, 0.04]
Subtotal (95% CI)			**42**			**45**	**40.8%**	**-1.05 [-1.66, -0.43]**
Heterogeneity: Tau² = 0.17; Chi² = 5.29, df = 3 (P = 0.15); I² = 43%								
Test for overall effect: Z = 3.33 (P = 0.0009)								
3.6.2 Glucose 4hr post-op								
Rimback 1986	4.5	0.8	11	6.8	0.3	11	7.3%	-3.66 [-5.12, -2.20]
Scott 1988	7.8	0.8	10	7.9	0.6	10	10.5%	-0.14 [-1.01, 0.74]
Wallin 1988	5.4	1.2	9	6.6	0.6	10	9.8%	-1.23 [-2.23, -0.23]
Subtotal (95% CI)			**30**			**31**	**27.7%**	**-1.59 [-3.39, 0.21]**
Heterogeneity: Tau² = 2.21; Chi² = 16.56, df = 2 (P = 0.0003); I² = 88%								
Test for overall effect: Z = 1.73 (P = 0.08)								
3.6.3 Glucose 8hr post-op								
Rimback 1986	6	0.3	11	5.9	0.3	11	10.7%	0.32 [-0.52, 1.16]
Scott 1988	7.7	0.5	10	7.9	0.5	10	10.5%	-0.38 [-1.27, 0.50]
Wallin 1988	6	0.3	9	6.1	0.5	10	10.4%	-0.23 [-1.13, 0.68]
Subtotal (95% CI)			**30**			**31**	**31.6%**	**-0.08 [-0.59, 0.43]**
Heterogeneity: Tau² = 0.00; Chi² = 1.42, df = 2 (P = 0.49); I² = 0%								
Test for overall effect: Z = 0.31 (P = 0.75)								
Total (95% CI)			**102**			**107**	**100.0%**	**-0.87 [-1.44, -0.30]**
Heterogeneity: Tau² = 0.61; Chi² = 32.78, df = 9 (P = 0.0001); I² = 73%								
Test for overall effect: Z = 2.98 (P = 0.003)								

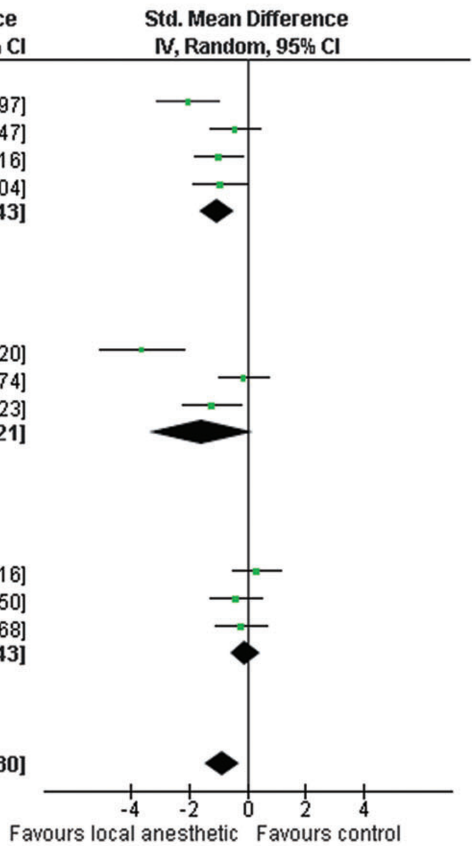

Fig. 4. Meta-analysis of the effect of intraperitoneal local anaesthetic on post-operative serum glucose in open general surgical procedures.

Study or Subgroup	Local anesthetic Mean	SD	Total	Control Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.4.1 Cortisol 1hr post-op								
Scott 1988	800	250	10	950	150	10	14.3%	-0.70 [-1.61, 0.21]
Tsuji 1983	548	180	12	716	153	14	15.1%	-0.98 [-1.80, -0.16]
Wallin 1988	650	130	9	625	180	10	14.4%	0.15 [-0.75, 1.05]
Subtotal (95% CI)			**31**			**34**	**43.8%**	**-0.52 [-1.19, 0.14]**
Heterogeneity: Tau² = 0.15; Chi² = 3.47, df = 2 (P = 0.18); I² = 42%								
Test for overall effect: Z = 1.54 (P = 0.12)								
3.4.2 Cortisol 4hrs post-op								
Scott 1988	1,300	300	10	950	200	10	13.6%	1.31 [0.33, 2.30]
Wallin 1988	890	150	9	710	190	10	13.8%	1.00 [0.03, 1.97]
Subtotal (95% CI)			**19**			**20**	**27.3%**	**1.15 [0.46, 1.84]**
Heterogeneity: Tau² = 0.00; Chi² = 0.20, df = 1 (P = 0.65); I² = 0%								
Test for overall effect: Z = 3.27 (P = 0.001)								
3.4.3 Cortisol 8hrs post-op								
Scott 1988	1,000	200	10	1,100	200	10	14.5%	-0.48 [-1.37, 0.41]
Wallin 1988	640	190	9	690	350	10	14.4%	-0.17 [-1.07, 0.74]
Subtotal (95% CI)			**19**			**20**	**28.9%**	**-0.32 [-0.96, 0.31]**
Heterogeneity: Tau² = 0.00; Chi² = 0.23, df = 1 (P = 0.63); I² = 0%								
Test for overall effect: Z = 1.00 (P = 0.32)								
Total (95% CI)			**69**			**74**	**100.0%**	**-0.00 [-0.63, 0.62]**
Heterogeneity: Tau² = 0.50; Chi² = 19.82, df = 6 (P = 0.003); I² = 70%								
Test for overall effect: Z = 0.01 (P = 0.99)								

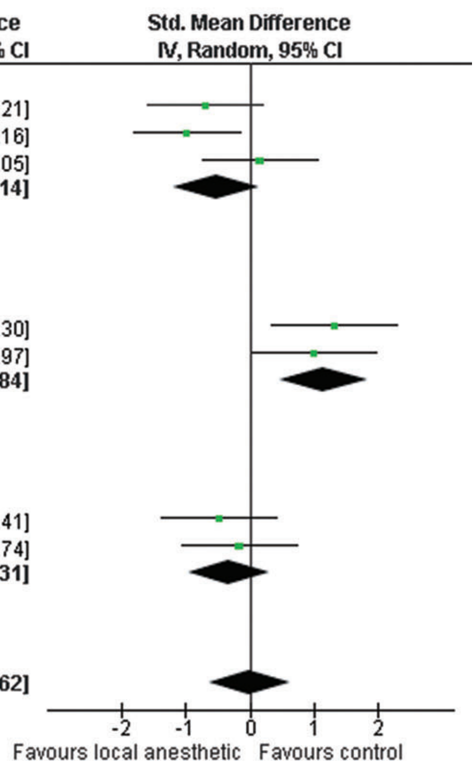

Fig. 5. Meta-analysis of the effect of intraperitoneal local anaesthetic on post-operative serum cortisol in open general surgical procedures.

is sufficient to achieve good visceral and peritoneal cover without putting the patient at additional risk of systemic toxicity if an infusion system were to be setup.

Complete 'noci-association' aims to use different approaches to silence all noxious stimuli that results from surgical manipulation of human tissue and forms the theoretical basis of preemptive analgesia.[31] The mechanisms that may possibly lead to quickened bowel recovery with IPLA may be as a result of noci-association and warrants further attention. It has been shown that intestinal motility is increased *in vitro* with LA blockade of discharge from the Auerbuch's plexus, thought to be due to blockade of intrinsic non-cholinergic mechanisms that normally exert a tonic inhibitory influence.[32,33]

Furthermore, intravenous LA induces a dose-dependent increase in rhythmic intestinal motility independent of background activity.[26] This may be due to blockade of inhibitory mechanisms described above or as a result of a direct excitatory influence on the viscera. Similar effects have also been shown *in vitro*, on intestinal muscle strips, suggesting that LA does have a direct effect on smooth muscle.[24,25]

LAs have well known anti-inflammatory actions. Animal work on IPLA has been interesting in that lignocaine and bupivacaine prevented peritonitis and adhesion formation by concentrated hydrochloric acid and eutectic mixture of local anesthetic (EMLA) (lidocaine/prilocaine) inhibited the adhesion formation after bacterial peritonitis in rats compared with controls.[22,23] Lignocaine applied to obstructed bowel serosa inhibited and reversed gut fluid losses and inflammation.[42] To our knowledge, these benefits have not been shown clinically.

The mechanism of action of IPLA is not fully understood, although it is likely that there is blockade of free afferent nerve endings in the peritoneum. Systemic absorption of LA from the peritoneal cavity may also play some part in reduced nociception. The peritoneum cavity is the largest serous membrane in the body and has a surface area of 1.8 m^2, close to that of the skin.[43] It is known that systemic levels of LA are detectable in the serum circulation as soon as 2 min after bolus instillation into the peritoneum, reaching a systemic maximum after 10–30 min.[44] Studies have confirmed that low dose intravenous LA infusion is advantageous in patients having abdominal operations with decreased pain, reduced opioid consumption and faster return of bowel function resulting in reduced hospital stay and intravenous LAs have been termed 'the poor man's epidural'.[45–48]

However, systemic absorption of LA after any route of administration occurs naturally such as after incision site injection, epidural infusion and transversus abdominus plane blockade. Whether all the effects of these methods can be attributed to systemic effects alone is very unlikely. IPLA will result in a local effect. However as discussed above, increasing the IP dose (which will result in increased systemic levels) does not necessarily translate into further clinical benefits.[40] This is seen routinely in patients with any infusion catheter blockade. Once an adequate block is obtained, increasing the infusion dose, does not translate into better pain control, while the systemic levels will be increased. There appears to be little difference in the pharmacodynamics of different agents, most reaching a maximum peak dose after 15–30 min after instillation.[49] However, adding adrenaline to the infusion significantly lowers the concentration maximum and prolongs the time to reach this maximum.

It is proposed that intravenous lignocaine may have a positive effect on gastrointestinal motility due to its opioid-sparing effect, prevention of inflammation and by decreasing inhibitory sympathetic tone. It is also known that LA's have anti-inflammatory actions.[50,51] Topical IP lignocaine and bupivacaine have been shown to inhibit chemical peritonitis in an animal model.[22] IP infusion of procaine has also been shown to reduce adhesion formation and density significantly in rabbits after abdominal surgery.[52]

Cervero argues that although visceral nociception is not easily excited in health, it may be increased in inflammation. Afferent signalling may therefore be greater in magnitude and prolonged in duration after injury such as surgical insult.[1] A pro-inflammatory cytokine cascade in the peritoneal cavity, with direct action on the visceral afferents and the vagus as a major vehicle, is a feasible contributor to post-operative visceral pain perception and the 'sickness response'.[53] Using field blockade with IPLA, it is plausible that peritoneal and visceral signalling to the brain maybe modulated, thereby possibly attenuating the impact of major abdominal surgery. Intra-colonic LA has been shown to effectively reduce visceral and cutaneous hypersensitivity clinically and in the lab.[54,55]

Four meta-analyses have previously been performed on the use of IPLA in laparoscopic cholecystectomy.[15,56–58] The main results in these meta-analyses are in limited favour of IPLA use. Trials utilizing IPLA in laparoscopy for indications other than cholecystectomy are emerging.[14,59,60]

This systematic review is limited by the quality of the included studies and the lack of standardized perioperative care between trials. The nature of clinical trials performed over 30 years meant that there was a variety of different drugs and methods of administration being used. Therefore, there is a fundamental heterogeneity in clinical studies that must be considered.

In conclusion, various methods of IPLA use have been described in open surgery. Meta-analyses showed reduced pain and attenuated hyperglycaemia responses to surgery, although patient numbers were small. There are reports of quickened return of bowel function in open surgery, although meta-analysis could not be performed for this outcome. There is room for further research in this area. No trials in colorectal and urological patients were identified in this review. Given the qualitative potential benefits discussed, in particular, that suggested on bowel function, future trials looking at this would further advance our understanding of this role of IPLA in the goal of complete noci-association. Furthermore, novel methods in delivery of agents to the peritoneal cavity are emerging with the advent of a peritoneal nebulizer, and the use of elastomeric pumps to maintain a continuous infusion system for IP drug delivery.[39,61–64] Future RCTs should utilize these technologies which enable an even and prolonged distribution of IPLA in already optimized settings.

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